Chapter 25. Independent Lung Ventilation

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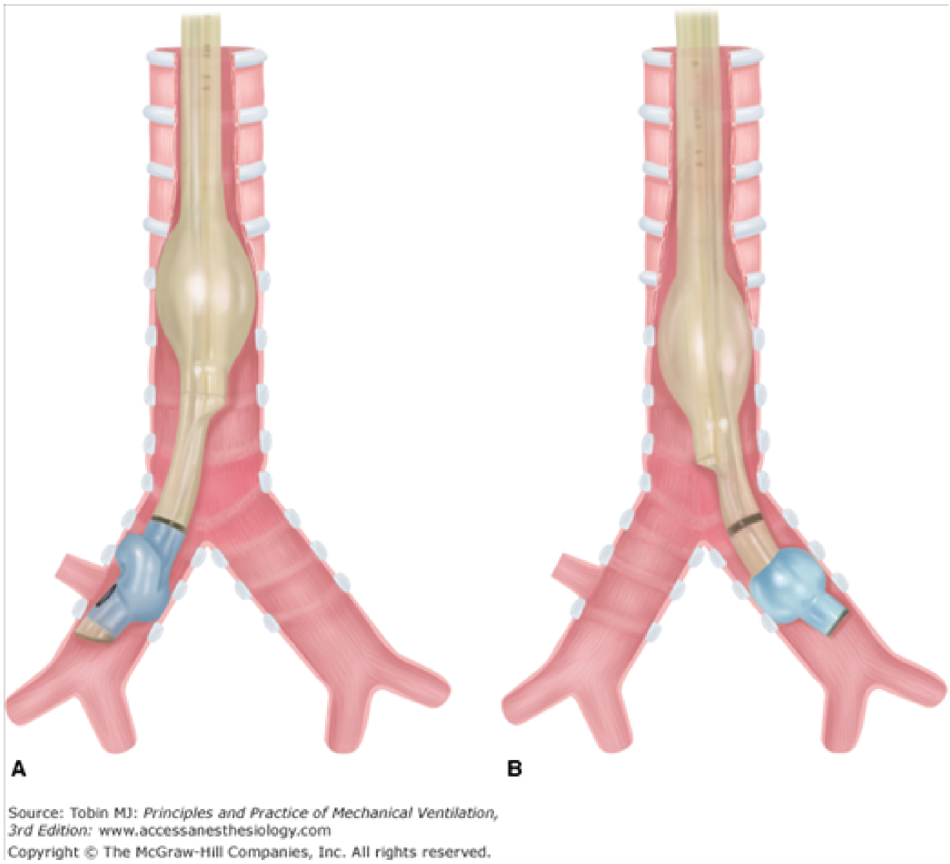
Independent Lung Ventilation: Introduction

Independent lung ventilation (ILV) was first used in thoracic surgery and the intubation devices were developed for this purpose. Gale and Waters first reported ILV in 1931, by passing a single-lumen endobronchial tube into the main bronchus of the dependent nonoperative lung for ventilation and exclusion of purulent secretion (if present) from the operative lung. In 1936, Magill[1] reported endobronchial placement of a suction catheter with a balloon to occlude the operative bronchus and tracheal placement of an ETT to ventilate the nonoperative lung. In 1947, Moody[2] encased the endobronchial balloon with metal studs to reduce the risk of balloon dislodgment.

The first double-lumen tube (DLT), enabling independent ventilation of both lungs, was reported by Carlens[3] in 1949. Thus, tube was similar to current left DLTs, but had a rubber “hook” to engage the carina for accurate placement. Although a major advance, this tube caused trauma and was unsuitable for left pneumonectomy. The first right DLT, which did not occlude the right upper lobe bronchus, was not reported until 1960 by White.[4] In 1962, Robertshaw[5] reported right and left DLTs, which served as the prototype of today’s DLTs. Current DLTs have replaced red rubber with polyvinyl chloride (PVC) to reduce mucosal injury and improve malleability and airflow (Fig. 25-1).

Figure 25-1



(**A**) Right and (**B**) left polyvinyl chloride (PVC) (Mallinckrodt) double-lumen tubes (DLT) shown against a schematic of the trachea and major airways.

For many years, DLTs and ILV were used entirely for thoracic surgery.[2,5,6] In 1976, Glass and Trew[7,8] and their coworkers reported ILV for nonsurgical purposes: respiratory insufficiency from unilateral lung disease. Since then, application of ILV has broadened to a wide range of conditions (Table 25-1) employing a variety of techniques. Institutions specializing in conditions commonly requiring ILV (such as single lung transplantation or alveolar proteinosis), ILV may be used in approximately 0.5% of all mechanically ventilated patients.[9,10] Although most intensive care units use ILV in fewer than one in 1000 patients requiring mechanical ventilation, it can be a lifesaving measure in specific conditions, making maintenance of suitable equipment and knowledge of its use required.

Table 25-1: Polyvinyl Chloride Double-Lumen Tubes: Choice of Size

Size (French)	OD (mm)	Tracheal ID (mm)	Bronchial ID (mm)	Use	Manufacturer[a]
26	8.7	3.5	3.5		Rüsch
28	9.3	3.1	3.2	Children weighing <40 kg	Mallinckrodt
32	10.7	3.5	3.4		Sheridan
35	11.7	4.5	4.3	Children weighing >40 kg	Mallinckrodt
37	12.3	4.7	4.5	Small adult	Mallinckrodt
39	13.0	4.9	4.9	Medium adults, usual female size	Mallinckrodt
41	13.7	5.4	5.4	Large adult, usual male size	Mallinckrodt

Abbreviations: *ID,* internal diameter; OD, outside diameter.

[a]Rüsch: Duluth, GA; Sheridan: Argyle, NY; Mallinckrodt: St. Louis, MO.

Adapted, with permission, from Campos JH.[267]

ILV is infrequently and often urgently required when conventional mechanical ventilation is not a viable alternative. As a result, there have been no randomized controlled trials of ILV in patients and current evidence is based almost entirely on animal models, case reports, and case series with before and after analyses.

Basic Principles of Independent Lung Ventilation

All the indications for ILV (see Table 25-1) are based on one or two fundamental requirements.

1. *The need to protect one lung from harmful effects of fluid in the other lung* (blood, purulent or malignant secretions, lavage fluid). Placing the fluid-filled lung in the dependent position (lateral decubitus) minimizes the risk of unwanted fluid entering the other lung but also maximizes hypoxia[11] by maximizing blood diversion to the less-functional lung. Placing the fluid-filled lung in the nondependent position improves oxygen saturation (SaO_2), but the risk from fluid spillage in this position is unacceptably high.[11] The supine position is usually the best compromise, except during thoracic surgery, where the operative lung must be in the uppermost position.
2. *The need to isolate the ventilatory patterns to each lung.* In each case, the ventilatory pattern for one lung has disadvantageous effects if applied to both lungs. This includes one-lung ventilation (OLV; e.g., thoracic surgery, whole-lung lavage), ventilation to each lung differing in only a single variable (e.g., positive end-expiratory pressure

[PEEP]), or ventilation requirements differing in every variable (e.g., single-lung transplant for obstructive airways diseases), or the need to prevent air loss from one lung (e.g., massive air leak).

If OLV is required, a bronchial blocker and endobronchial tube or a DLT may be used. If ILV is required, then a DLT is usually used, although some bronchial blockers can allow limited ventilation such as continuous positive airway pressure (CPAP) or jet ventilation.

Irrespective of technique, tube or blocker position is usually checked using fiberoptic bronchoscopy immediately after instigation, and lung isolation is leak tested. This is important for all indications for ILV but is most critical where protection from secretions is required.

Independent Lung Ventilation Techniques

Lung Separation

ILV must be preceded by the placement of a DLT or alternate lung isolation device. These must be introduced, correctly positioned, and then tested to ensure isolation of lung ventilation.

Compared with red-rubber DLTs,[3,4,12] PVC-type DLTs (see Fig. 25-1, e.g., Mallinckrodt, Sheridan, Rusch, Concord, Portex, Marraro) are more flexible, have better internal–external diameter[10] ratios, better gas-flow characteristics, easier suction and bronchoscopy access,[13] allow airway seal with lower cuff pressures,[14] are less irritating to respiratory mucosa, have a lower risk of trauma, and are easier and quicker to position.[15] These characteristics make PVC types the DLTs of choice, although airway injury from PVC DLTs may still occur.[16,17]

For most indications, a left DLT (see Fig. 25-1) should be used because placement is easier than for a right DLT, which has a high risk of right upper-lobe occlusion.[18] A right DLT (see Fig. 25-1) is required for thoracic surgical procedures that include the left main bronchus (left pneumonectomy, left main bronchial lesions, stenosis, or rupture), thoracic aortic aneurysm repair, or anatomic abnormalities that prevent satisfactory access to the left main bronchus.[19–21] A right DLT is also required for ILV with a recent left single lung transplantation to avoid injury to the anastomosis and distal airway. To minimize airflow resistance and maximize endobronchial access, the largest DLT that will not cause laryngeal or airway injury should be chosen (see Table 25-1).

PVC DLTs are usually introduced with the endobronchial curvature angled anteriorly and a rigid stylet in situ to facilitate the passage of the tip through the vocal cords. Once through the cords, the stylet is usually removed and the DLT is rotated through 90 degrees so that the endobronchial curvature is directed toward the appropriate side. The tube is then advanced until an increase in resistance is detected. In a randomized trial of sixty patients receiving a DLT for thoracic surgery, Lieberman et al[22] compared removal of the stylet (as above) and leaving the stylet in situ until placement was complete; the latter increased the correct placement rate from 17% to 60%.

DLT tube position and function must then be confirmed by one or more of three techniques:

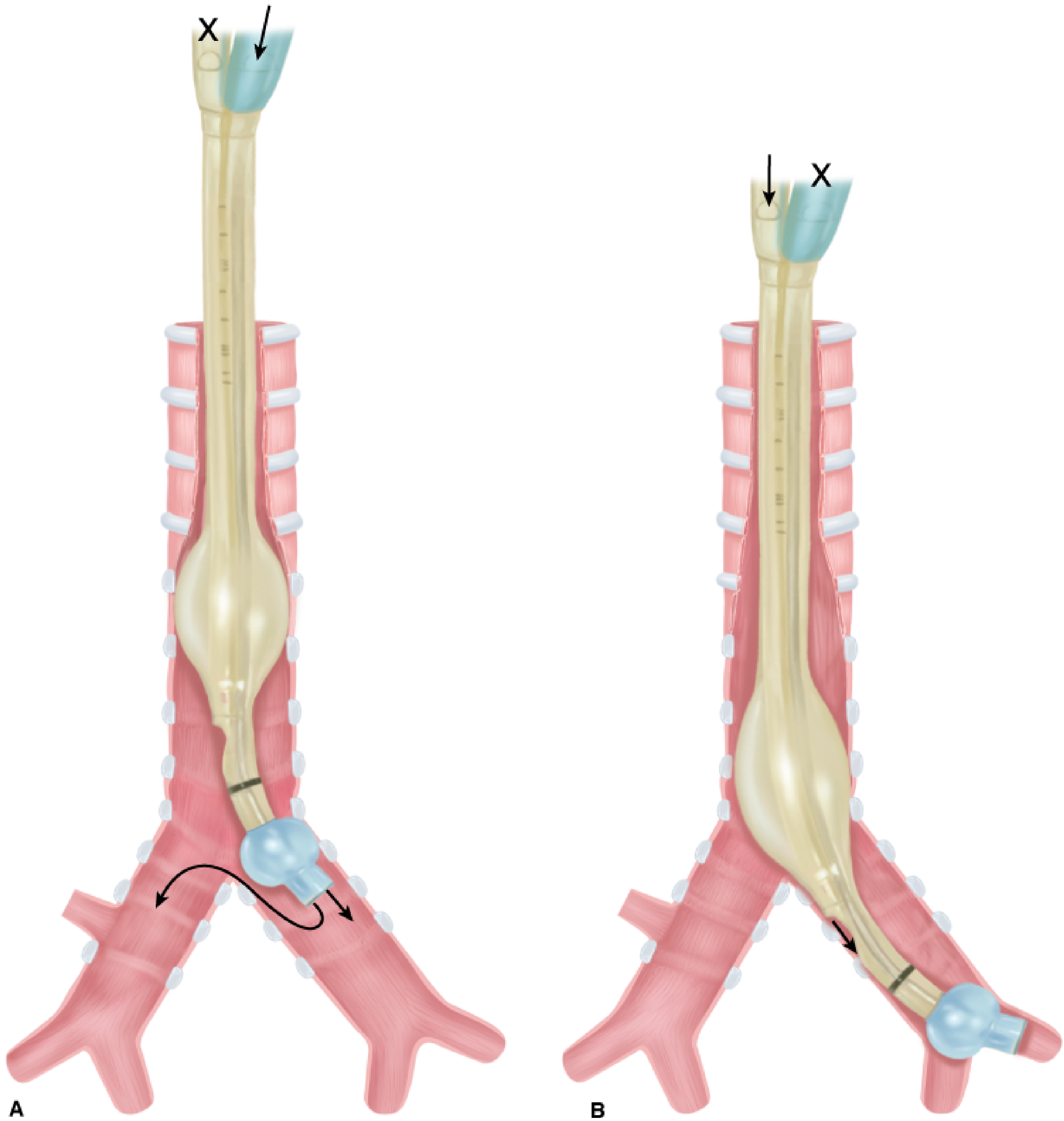
1. **Auscultation.** Following cuff inflation, the tracheal port should be clamped and the bronchial port ventilated.[19] Bilateral or contralateral breath sounds indicate placement is too proximal and a need to reposition the DLT (Fig. 25-2A), whereas breath sounds heard only on the correct side indicate correct placement (see Fig. 25-1). Difficulty with ventilation should be resolved by deflating the bronchial cuff: bilateral breath sounds indicate that placement is too

proximal, whereas breath sounds heard only on the side of the endobronchial tube indicate that placement is too distal (Fig. 25-2B). Lung auscultation should always include both upper and lower lobes, to ensure correct placement of the endobronchial port within the bronchus, especially with a right DLT where the right upper lobe is easily occluded.

2. **Bronchoscopy.** Following DLT insertion, a small fiberoptic bronchoscope may be passed through the tracheal lumen to confirm position of the tracheal port and that the endobronchial tube is in the correct position.[19,23] The endobronchial cuff should be visible just distal to the carina. Subsequent insertion of the bronchoscope down the endobronchial lumen may be used to confirm that endobronchial placement is not too distal and that, with right DLTs, the right upper-lobe bronchus is over the corresponding fenestration. Although not essential for DLT placement, bronchoscopy reliably confirms accurate placement and should be used routinely. It has an advantage with anatomic variations and in detecting partial airway occlusions.[17]

3. **Leak test.** While ventilating one lung, a connection to the second airway can be placed under water. Any bubbling indicates air leak from the ventilated lung to the opposite side. Such leaks may not be detected by auscultation or bronchoscopy and may be important, particularly when one of the goals is to protect one lung against fluid (whole-lung lavage, blood, or purulent secretions) from the other lung. Bubbling may indicate the need for tube repositioning or higher bronchial-cuff inflation.

Figure 25-2



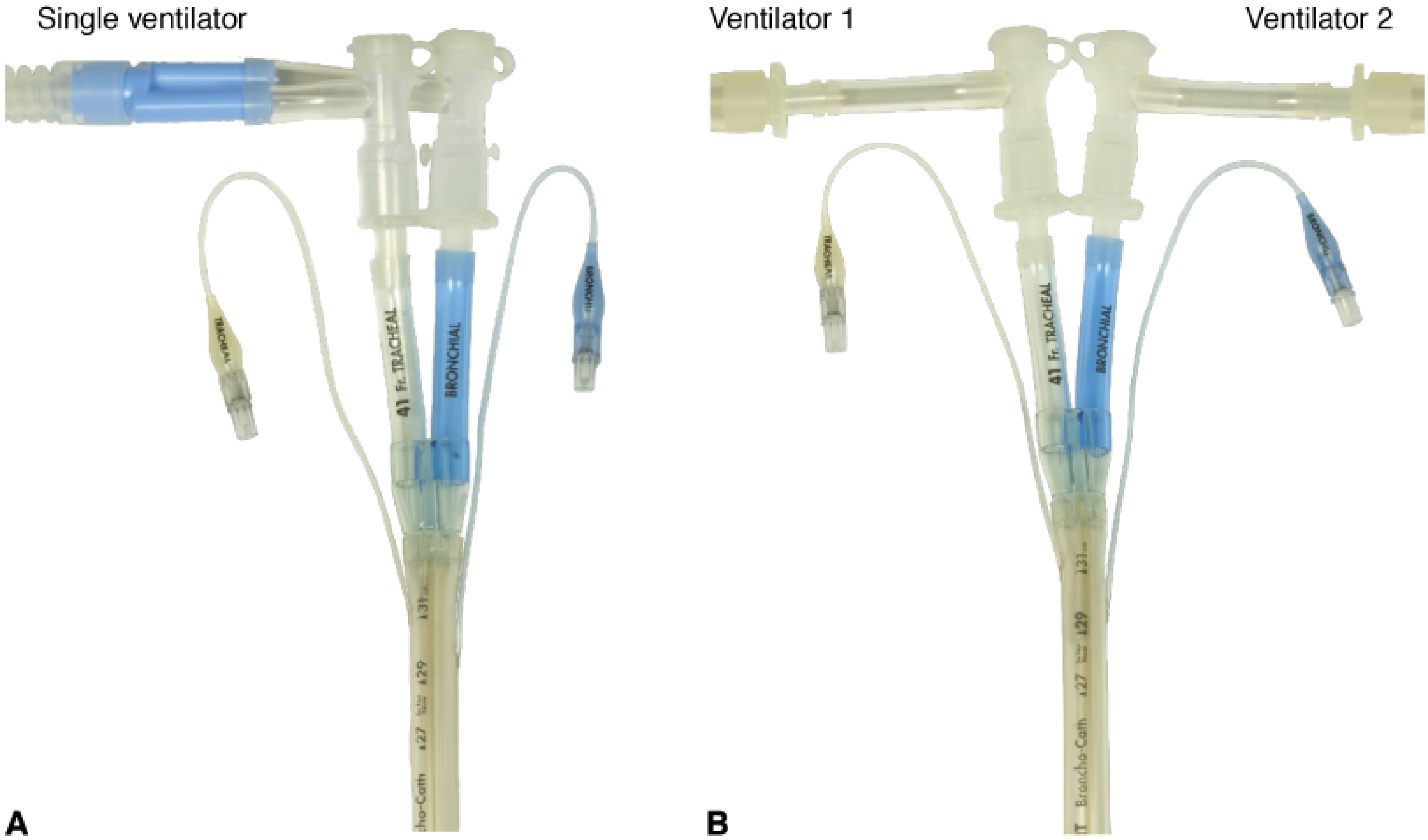
Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation, 3rd Edition:* www.accessanesthesiology.com

Flow patterns that may occur during auscultatory verification of a left DLT position (**A**) when DLT is insufficiently inserted and the left lumen is ventilated (right lumen clamped) and (**B**) when DLT is inserted too far and the right lumen is ventilated (left lumen clamped).

Chest radiography may be used to visualize DLT position, but is insufficient to verify critical tube placement or functional isolation.

DLTs may be connected to a single ventilator (e.g., for hemoptysis or at the beginning and end of whole-lung lavage; Fig. 25-3A) or connected to two separate ventilators (see Fig. 25-3B).

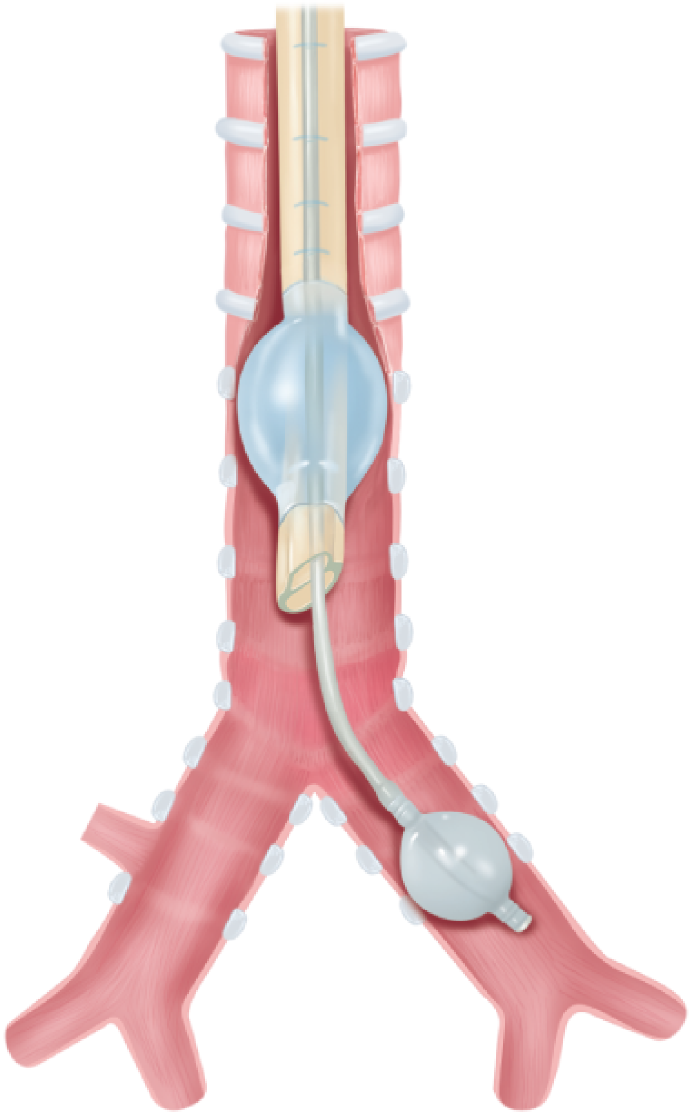
Figure 25-3



A. Both lumens of a double-lumen tube connected to a single ventilator airway. **B.** Each lumen of a double-lumen tube connected to separate ventilators.

Bronchial blocking techniques and selective endobronchial intubation are alternatives to a DLT. The right or left main bronchus may be blocked by placement of a balloon-tipped catheter into that bronchus. This allows OLV and is suitable for thoracic surgery, bleeding, or fistula control. The catheter (Arndt endobronchial blocker, Magill blocker, Cohen Flexitip Endobronchial Blocker, Fogarty or Foley catheter)[24–28] may be placed outside or within a standard cuffed endotracheal tube (ETT) lumen or passed down a specially designed second small lumen in the ETT (Univent tube).[29–31] A Univent tube (Fig. 25-4) has a coudé-tipped bronchial blocker that allows blind guidance of the blocker into the desired bronchus with auscultatory confirmation of correct positioning. Bronchoscopic confirmation of best position within the airway, however, is still recommended. In addition, the Univent's axial-blocker shaft has a lumen that allows irrigation, suction, O_2 insufflation, CPAP, and high-frequency ventilation.[29] The Arndt endobronchial blocker[24–26] (Fig. 25-5) requires bronchoscopic guidance for placement. The kit comes with an ETT adaptor that allows access for mechanical ventilation, the blocker, and the bronchoscope through separate ports (Fig. 25-5). The bronchoscope is passed into the airway that requires blocking, and the blocker is then guided into that airway via a snare over the bronchoscope (Fig. 25-5A). The bronchoscope then is withdrawn, the balloon is inflated, and its position is confirmed by bronchoscopy before withdrawal (Fig. 25-5B). This has the advantage of being performed via the ETT in situ, thereby avoiding reintubation, provided that the ETT is sufficiently large to admit both the blocker and the available bronchoscope. The Cohen Flexitip Endobronchial Blocker[28] has a flexible tip that can be guided under bronchoscopy but independently of the bronchoscope (Fig. 25-6).

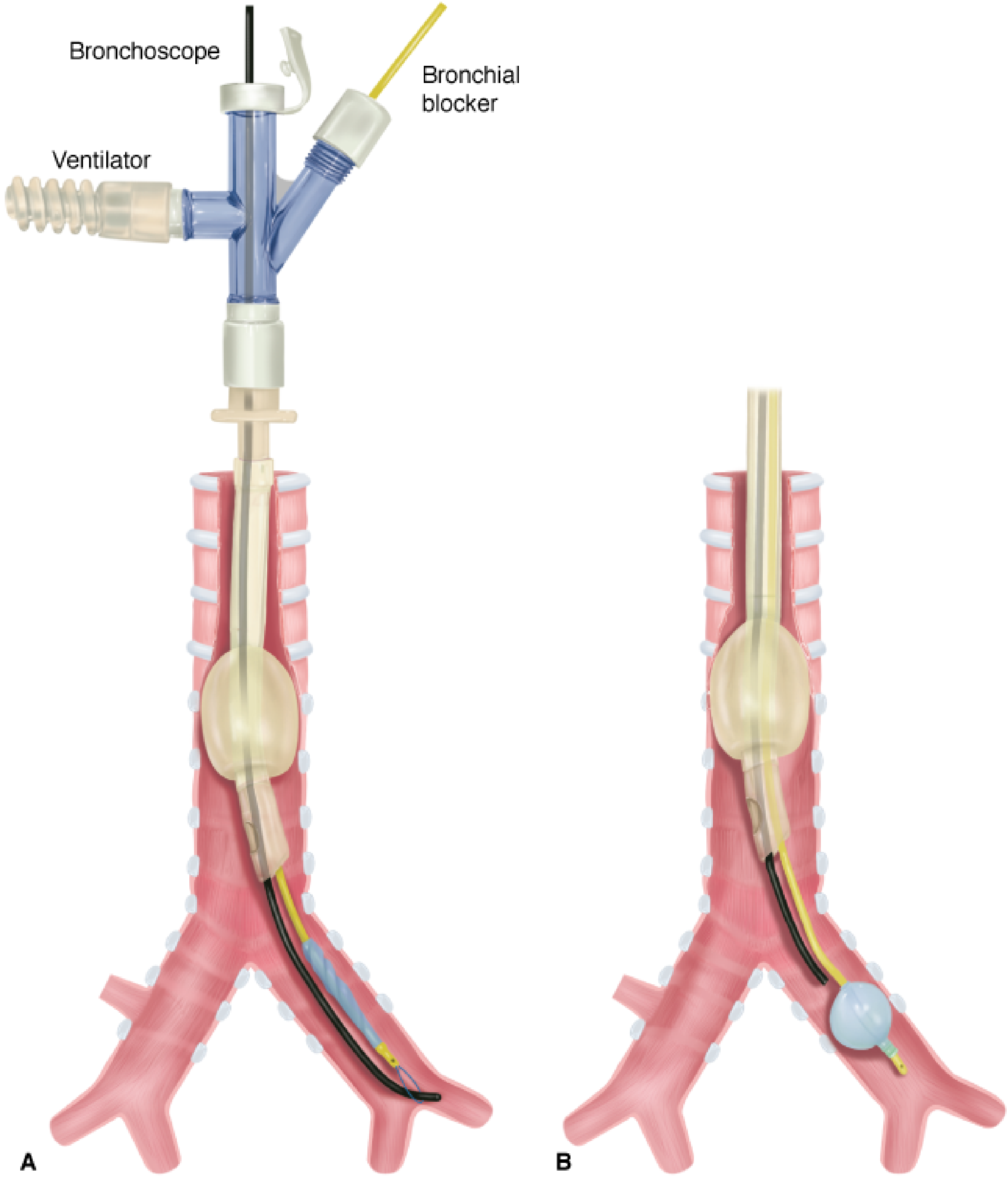
Figure 25-4



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation, 3rd Edition:* www.accessanesthesiology.com

The Univent tube with the balloon inflated in the left main bronchus.

Figure 25-5



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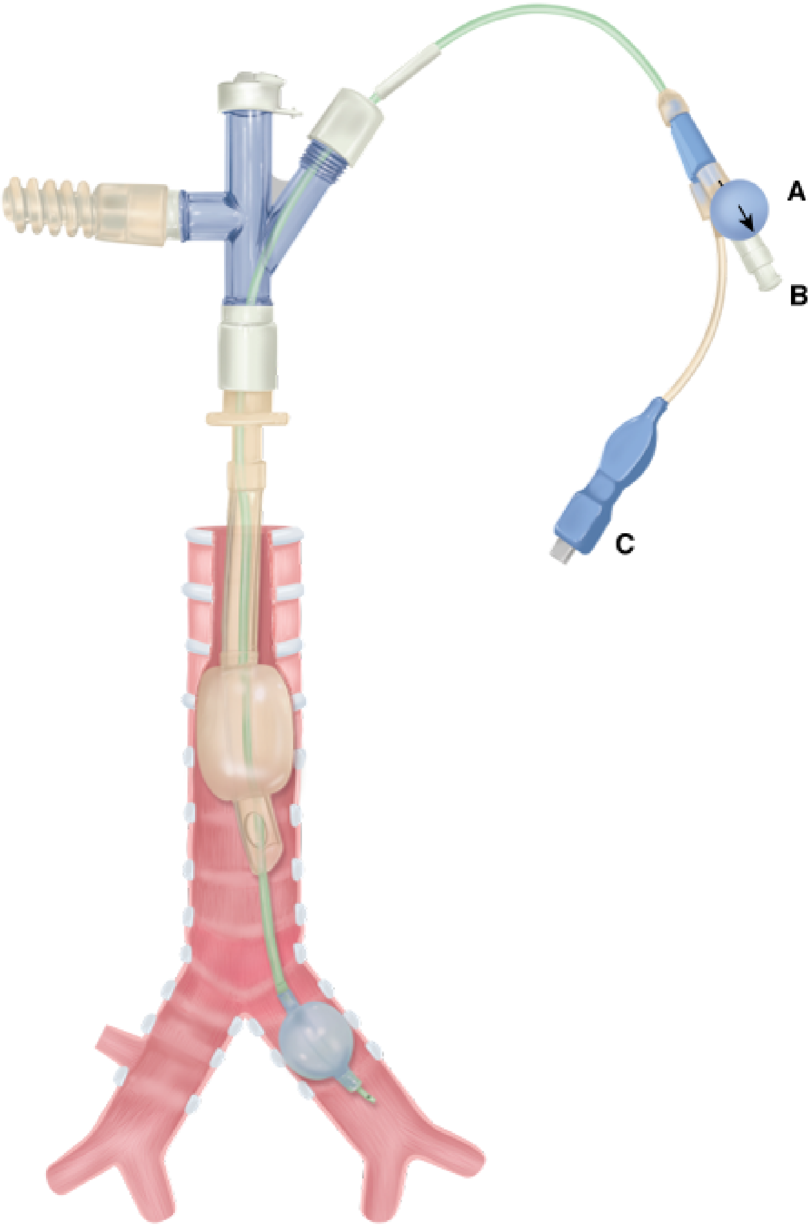
The Arndt endobronchial blocker shown (**A**) during bronchoscope guidance into the left main bronchus and (**B**) after partial bronchoscope withdrawal and balloon inflation, with the bronchoscope remaining to check balloon position.

Figure 25-6



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The Cohen Flexitip Endobronchial Blocker shown with its endotracheal tube attachment. **A** is the knob that controls and locks the tip flexion. **B** is a Luer lock port to the lumen of the Cohen blocker. **C** is the valved insufflation port for the blocker balloon.

Techniques for Independent Lung Ventilation

A variety of techniques have been reported.

Synchronized Independent Lung Ventilation

Synchronized independent lung ventilation (SILV) consists of synchronous initiation of inspiration into each lung. Each lung must necessarily have the same respiratory rate, but may have different tidal volume (V_T), PEEP, and inspiratory flow. SILV may be achieved by a variety of techniques.

1. *Two ventilators of the same type linked to cycle synchronously.* This may be achieved by electronically "slaving" the rate control of a second ventilator to a primary ("master") ventilator,[18,32–40] by electronically synchronizing both to an external control device,[41,42] or by simultaneously resetting the respiratory cycle on paired ventilators and relying on accurate internal timing to maintain synchronization.[43,44] These forms of SILV allow difference in all variables apart from rate—different V_T, PEEP, and inspiratory flow, and hence different inspiratory-to-expiratory time (T_I/T_E) combinations. SILV with two ventilators synchronized 180 degrees out of phase[45,46] has been successful in animals, but appears to have no advantages over other forms of ILV and has not been reported in humans.

2. *A single ventilator linked to a twin circuit with variable resistances in each inspiratory line*[47–49] can create different flows to each lung. The V_T received by each lung will then be determined by both the resistance in the circuit and the impedance in the lung and must be independently measured in each circuit and resistance adjusted accordingly. An alternative to this is flow controllers in both inspiratory lines[50,51] resulting in a fixed V_T to each lung determined by the set flow and inspiratory time and independent of lung impedance. Separate PEEP in each circuit was achieved by expiratory flow controllers. This method allows different V_T and PEEP to each lung, but necessarily must have the same T_I, T_E, and rate.

3. *A single ventilator linked to two circuits, each with a separate PEEP valve or other PEEP-generating device.*[52,53] This arrangement allows different PEEP to each lung. The division of V_T between the lungs is not controlled independently, being determined by both the relative inherent impedance of each lung and the effect of PEEP on that impedance.

4. *A single ventilator linked to two circuits with no attempt to influence the distribution of ventilation.*[13] This method generally is used during selective airway protection. The division of V_T between the two lungs is determined solely by their relative impedance.

While many indications for ILV are suited to having the same ventilator rate to each lung, there is usually no particular benefit for exact coordination of two ventilators. The only exception may be the uncommon circumstance where patient-triggered ventilation is attempted.

Asynchronous Independent Lung Ventilation

Asynchronous independent lung ventilation (AILV) consists of completely independent ventilator techniques applied to each lung. It requires two separate ventilation devices. Options include:

1. *Controlled* (CMV) *or intermittent mechanical ventilation to both lungs.*[9,54–62]

2. *CMV or intermittent mechanical ventilation to one lung and high-frequency jet ventilation* (HFJV) *to the other.*[57,63–66]

3. *CMV or intermittent mechanical ventilation to one lung and CPAP to the other.*[7,57,58,67]

4. *High-frequency oscillatory ventilation to both lungs.*[68]

AILV permits different rate, V_T, inspiratory flow, and PEEP to each lung. Lack of synchronization between the two lungs offers the greatest flexibility and appears to hold no disadvantage and a number of advantages compared with SILV.[55,61]

One-Lung Ventilation

With OLV, one lung is ventilated mechanically while the other is either occluded or open to atmosphere with an option of spontaneous ventilation. OLV is used mainly in thoracic surgery to keep the lung collapsed and immobile. It also may be used in bronchopleural fistula (BPF) to prevent air leak or for selective airway protection if secretions from the affected lung prevent any useful ventilation (e.g., massive hemoptysis). Options include:

1. *DLT intubation*[11,20,21,69–71] or in infants, two separate uncuffed tracheal tubes of different lengths attached longitudinally (Marraro DLT)[69] with CMV applied to only one lumen.
2. *Endotracheal intubation with a bronchial blocker inserted into one of the major bronchi.* The occlusive balloon excludes that lung from mechanical ventilation and the blocker lumen allows deflation (and inflation) of that lung.[1,21,24–26,30]
3. *Endobronchial intubation with a single lumen tube* (e.g., Mackintosh-Leatherdale left endobronchial tube and Gordon Green right endobronchial tube).[21]

Spontaneous Ventilation

Spontaneous ventilation with a DLT and differential CPAP applied to each lung was first reported by Venus et al.[72] It has been used in spontaneously breathing patients with unilateral pulmonary contusion and atelectasis with good results.[72,73] It is not recommended because of increased work of breathing through a long narrow tube.

Applications of Independent Lung Ventilation

Applications of ILV (Table 25-2) fall into five categories.

Table 25-2: Indications for Independent Lung Ventilation

Thoracic Surgery
Pneumonectomy
Some lobectomies
Thoracic aortic surgery
Thoracoscopy
Some esophageal surgery and diaphragmatic surgery
Selective airway protection
Secretions (tuberculosis, bronchiectasis, abscess)
Whole-lung lavage
Massive hemoptysis
Bronchial repair protection
Bronchopleural fistula
Asymmetrical lung disease
Unilateral parenchymal injury
Aspiration
Pulmonary contusion
Pneumonia
Massive pulmonary embolism
Reperfusion edema
Asymmetrical acute respiratory distress syndrome
Asymmetrical pulmonary edema
Atelectasis
Unilateral airflow obstruction
Single lung transplantation for chronic airway obstruction
Unilateral bronchospasm
Severe bilateral lung disease
Acute respiratory distress syndrome
Aspiration
Pneumonia

Thoracic Surgery

A number of thoracic surgical procedures require OLV, usually in the lateral position. Table 25-1 lists the common thoracic surgical indications for OLV.[19,21,69] Although OLV is required during surgery, ILV may be required before, and sometimes after, the surgical procedure.[69,74]

OLV may be achieved with a DLT, bronchial blocker, or endobronchial tube. A DLT has the advantage of enabling ILV, which may be important in alleviating hypoxia, but has the disadvantage of high-resistance airways with more difficult suctioning. An ETT with a bronchial blocker has the advantage of a wide-bore, low-resistance endotracheal lumen with better bronchoscopic and suction access, through which either OLV (bronchial blocker inflated) or double-lung ventilation (bronchial blocker deflated) can be delivered.[30] The narrow lumen in some bronchial blockers can enable lung inflation, deflation, CPAP, or HFJV, but does not allow conventional ILV. OLV may be delivered to either lung, but repositioning of the bronchial blocker is required.[30] When the blocker is an integral part of the ETT (Univent tube, see Fig. 25-4),[29–31] it is easier to insert and it maintains a more stable position compared with an intraluminal or extraluminal blocker that is more subject to movement and dislodgment, especially during suctioning, bronchoscopy, or patient movement.[30] With any bronchial blocker, bronchoscopy is easier and required less frequently.[19,30] A disadvantage is that the inflated blocking balloon near the site of intended bronchial surgery (e.g., pneumonectomy, single lung transplantation) may hamper that procedure, and a DLT placed in the opposite lung may be technically easier.

Effects of Independent Lung Ventilation in the Lateral Decubitus Position

Irrespective of patient position, gravity-induced differences in ventilation and perfusion occur vertically within each lung: between the bases and apices in the erect position, and between the dependent and nondependent lung regions in the supine position.

When a patient is in the erect or supine position, gravitational differences in ventilation and perfusion occur vertically within each lung. In the lateral decubitus position, these gravitational differences occur between the two lungs.[71,75] Up to two-thirds of perfusion shifts to the dependent lung.[75–77] During *spontaneous ventilation,* a smaller increase in ventilation also occurs in the dependent lung in the lateral position, thereby reducing ventilation–perfusion mismatch.[71,75] When the patient is anesthetized and *mechanically ventilated* in the lateral decubitus position, however, reduced diaphragmatic activity allows the weight of abdominal contents to retard expansion of the dependent lung,[71,75] and most ventilation is diverted to the nondependent lung. In this situation, the nondependent lung may receive up to two-thirds of ventilation.[36,40,76] Opening the nondependent thorax further reduces ventilation to the dependent lung by facilitating expansion of the nondependent lung.[77,78] The net effect of these changes is overperfusion relative to ventilation (or shunting) in the dependent lung and overventilation relative to perfusion (or dead space ventilation) in the nondependent lung with adverse effects on gas exchange.

Left thoracotomy achieves a higher partial pressure of arterial oxygen (Pa_{O_2}) during OLV than does right thoracotomy because the left lung normally receives 10% less cardiac output than the right lung.[70] The presence of chronic airway obstruction (CAO) is associated with better Pa_{O_2} during OLV possibly because of dynamic hyperinflation in the dependent lung.[70] Pa_{O_2} during double-lung ventilation is also predictive of during Pa_{O_2} OLV.

Some forms of ILV cannot be applied when the nondependent lung is immobile, deflated, or removed, yet ILV has been shown to improve many of these abnormalities. Application of PEEP to both lungs improves dependent-lung ventilation and oxygenation.[76] SILV (see "Techniques for Independent Lung Ventilation") with delivery of equal V_T to both lungs,[39] or PEEP to the dependent lung,[37,38,40,79,80] or both,[38] improves oxygenation when compared with double-lung ventilation or generalized PEEP.

When OLV is undertaken and the nondependent lung is excluded from ventilation, the reduction in ventilation to the dependent lung is eliminated, but all perfusion through the nondependent lung constitutes a shunt. In this situation, the amount of perfusion of the nondependent lung is a major determinant of hypoxia. Hypoxic vasoconstriction (not opposed by anesthetic agents), lung collapse, and surgical occlusion of blood flow (pneumonectomy) to the nondependent lung all have the potential to reduce shunt and improve Pa_{O_2} in OLV.

PEEP in the dependent lung may be beneficial by increasing functional residual capacity and improving distribution of ventilation or be detrimental by increasing alveolar pressure and diverting blood flow to the nonventilated lung.[71,81–86] Consequently, PEEP to the dependent lung during OLV may improve Pa_{O_2},[81,86] decrease Pa_{O_2},[81,82,86] or cause no change.[84–86] During OLV, Cohen et al[79] compared PEEP (10 cm H_2O) to the dependent lung, CPAP (10 cm H_2O) to the nondependent lung, and both. All three maneuvers increased compared with OLV alone. PEEP caused the smallest (nonsignificant) increase in Pa_{O_2}. CPAP and PEEP + CPAP caused larger (significant) increases in Pa_{O_2}, whereas PEEP + CPAP reduced cardiac output and oxygen (O_2) delivery significantly, which CPAP alone did not. Cohen et al[28] concluded that CPAP alone (to the nondependent lung) had the most beneficial effect by diverting blood flow to the dependent lung without reducing cardiac output. More recently, PEEP has been shown to benefit oxygenation during OLV to the dependent lung during thoracotomy.[87]

Because of these variable effects, it has been recommended that CPAP (5 to 10 cm H_2O) be applied to the nondependent lung, combined with no, low, or high PEEP (5 to 15 cm H_2O) to the dependent lung, depending on patient response.

Although function of the two lungs commonly differs after thoracotomy, conventional mechanical ventilation or spontaneous ventilation are usually resumed after surgery. ILV is required occasionally in the postoperative period for marked asymmetry of lung function with hypoxia,[7,8,55] BPF,[47,58,63,66,88] or following esophagectomy.[74] Pawar et al[69] reported successful use of the Marraro pediatric double-lumen tube[89] (two separate uncuffed tracheal tubes of different lengths attached longitudinally) in seventeen children, ages 1 day to 3 years (2.7 to 12 kg) who required OLV during cardiothoracic surgery. Of these, six children required ILV up to 48 hours postoperatively. All children survived. Ito et al[90] reported ILV with a separate bronchial cannula in each main bronchus, combined with bilateral HFJV, the repair of tracheal stenosis in a 2-year-old infant.

Selective Airway Protection

Whole-Lung Lavage

Pulmonary alveolar proteinosis is the most common indication for whole-lung lavage. This condition was first described in 1958,[91] and soon after, bronchopulmonary lavage became established as the main treatment.[92–100]

Pulmonary alveolar proteinosis is most often acquired (90%) but may be congenital or secondary to conditions such as acute silicosis and other inhalational syndromes, immunodeficiency disorders, malignancies, hematopoietic disorders, and lysinuric protein intolerance.[101] It has also been reported following prolonged cotton dust exposure.[102] Recent studies suggest that acquired pulmonary alveolar proteinosis may be caused by deficiency of granulocyte-macrophage colony-stimulating factor,[101,103] and administration of this factor is of value in approximately 50% of patients. Despite these advances, lavage remains the most effective therapy.[101,103]

The clinical course is variable: increasing dyspnea progressing to respiratory insufficiency, static disease with minimal symptoms, asymptomatic disease, and spontaneous resolution in some patients.[101,104] Because the course is variable, the decision to undertake bronchopulmonary lavage is based on disease progression and symptoms.

Whole-lung lavage has been used for asthma,[105,106] chronic bronchitis, and cystic fibrosis,[11,98,105–111] but with doubtful benefit; it is no longer recommended. Radioactive dust inhalation is another indication.[112]

Procedure for Whole-Lung Lavage.

To minimize hypoxia during the procedure, the worst lung should be lavaged first. The worst lung can be identified by chest radiograph, ventilation–perfusion scan, or oxygenation[3,113,114] during OLV to each lung before the procedure.

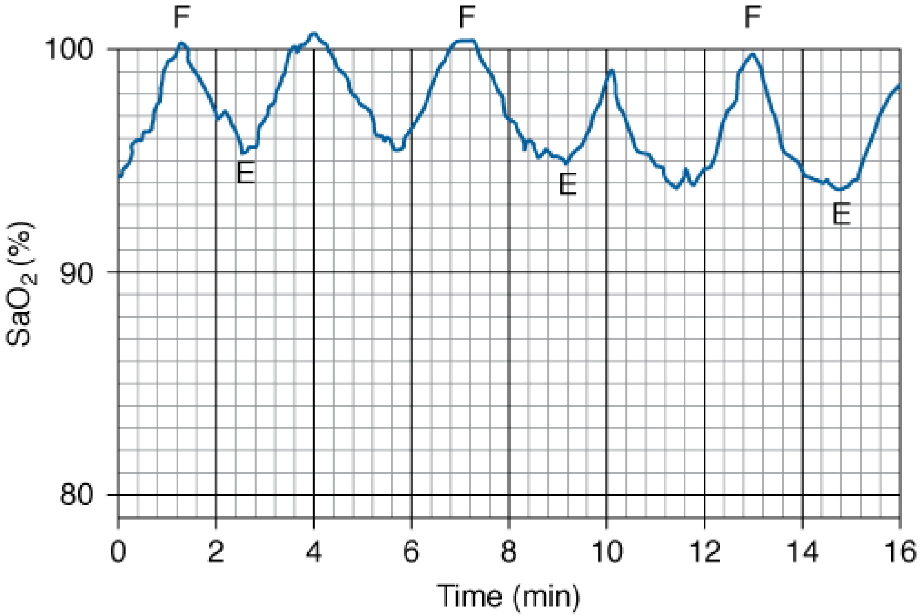
The procedure usually is performed with anesthesia, neuromuscular paralysis,[11,105,115] and a left DLT in the supine position. The supine position is chosen to balance the risk between hypoxia and fluid spillage[11] (see “Basic Principles of Independent Lung Ventilation”).

Complete lung isolation to prevent spillage of fluid from the lavaged lung into the ventilated lung is essential. Correct DLT position should be established using both bronchoscopy and leak testing (as above) up to plateau pressures of 40 to 50 cm H_2O.[11,19] Both lungs should be preoxygenated with 100% O_2 to maximize gas exchange and eliminate nitrogen, which may prevent full access of lavage fluid to the lung to be lavaged.

Isotonic saline, warmed to body temperature, then should be infused through wide-bore tubing into the lung to be lavaged from a gravitational height 30 cm above the midaxillary line.[11] This infusion pressure of 30 cm H_2O usually results in infusion volumes of 500 to 1000 mL depending on the compliance of the lung. Efflux of fluid may be commenced as soon as fluid influx is complete; the drainage tube is placed below the patient, assisted by head-down posturing, and percussion and vibration of the hemithorax.[11,115] Fluid flow may be achieved using separate clamped inlet and outlet lines or a single line that is elevated for fluid influx and lowered for fluid efflux. Total fluid exchange may range from 10 to 50 L and should be continued until efflux fluid is relatively clear.

When fluid influx into the lavaged lung is complete, the alveolar pressure usually exceeds the pulmonary capillary pressure, minimizing shunt through this lung and maximizing arterial oxygenation at this stage of the procedure.[105,115–118] When the lavaged lung is emptied of fluid, blood flow returns, and significant hypoxemia can occur (Fig. 25-7).[11,105,115–117]

Figure 25-7



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Variations in arterial oxygen saturation (SaO_2) during the influx (**F**) and efflux (**E**) phases of whole-lung lavage. (Modified, with permission, from Claypool et al.[115])

On conclusion of the lavage, double-lung ventilation should be recommenced with the DLT in situ using a single ventilator and a bifurcated circuit. If oxygenation is satisfactory, the patient may be weaned and extubated or reintubated with a regular ETT and later weaned. If oxygenation is unsatisfactory, ILV may need to be recommenced. Up to 1000 mL of saline may be retained in the lavaged lung;[115] although this is absorbed rapidly, lung function may not improve for hours or days. The procedure is repeated on the second lung after an interval of 1 to 3 days.

Leakage of fluid into the ventilated lung may be recognized by desaturation, crepitations, and rhonchi in the ventilated lung; fluid in the tubing to the ventilated lung; or air bubbles in the fluid from the lavaged lung.[11] If this occurs, lavage should be ceased, and the patient should be placed in the lateral decubitus position with the lavaged lung dependent and the head down to facilitate drainage from both lungs. Active suctioning of both lungs should be undertaken. If the leak is only minor and adequate oxygenation is restored, the correct DLT position may be reestablished, lung isolation rechecked, and the procedure continued. With a major leak and failure to restore adequate oxygenation once fluid removal is complete, double-lung ventilation with PEEP should be resumed and further lavage delayed until hypoxia has improved.

Hypoxemia during the procedure is common. Some patients with severe lung disease are too hypoxemic before the procedure to tolerate OLV. Several options exist for such patients. Extracorporeal membrane oxygenation (ECMO) may be instituted before lavaging the first lung.[11,119–123] ECMO may or may not be required for lavage of the second lung, depending on the effect of the first lung lavage on subsequent oxygenation. Cohen et al[121] reported successful lavage of both lungs with ECMO during a single session, avoiding the need for a second procedure. Because of the complexity and limited availability of ECMO, an alternative is to perform limited bronchoalveolar lavage using a fiberoptic bronchoscope,[124,125] with or without an inflatable cuff on the bronchoscope, under local anesthesia. This may be undertaken in a spontaneously breathing patient or during CMV through a single-lumen ETT. Limiting lavage to a lobe or several subsegments necessitates multiple procedures but minimizes hypoxemia. Nadeau et al[126] reported that the combination of inhaled nitric oxide and inflation of a balloon in the right pulmonary artery during right whole-lung lavage improved oxygenation sufficiently to avoid ECMO.

Randomized studies of whole-lung lavage for pulmonary alveolar proteinosis have not been undertaken. Seymour and Presneill[101] analyzed survival of 231 patients in multiple reports; 5-year actuarial survival from diagnosis was 94% ± 2% in patients who underwent whole-lung lavage (*n* = 146) compared with 85% ± 5% for patients who did not (*n* = 85). In fifty-five patients in whom duration of benefit was reported,[101] median duration of benefit from lavage was 15 months; fewer than 20% of patients followed beyond 3 years remained symptom-free. In a series of twenty-one patients followed prospectively after whole-lung lavage, Beccaria et al[127] found that more than 70% remained free from recurrent pulmonary alveolar proteinosis at 7 years. Although whole-lung lavage is labor-intensive, these data suggest that it is worthwhile.

Massive Hemoptysis

Massive hemoptysis carries a high mortality[128–130] and requires prompt intervention. Common causes include tuberculosis, bronchiectasis, lung abscess, mycetoma, pulmonary carcinoma, cystic fibrosis, arteriovenous malformations, and trauma.[104,128,131–133] The source of bleeding is the bronchial arterial system in 90% of patients.[134] Iatrogenic causes are uncommon but include pulmonary artery rupture by a pulmonary artery catheter[128,135,136] and transbronchial lung biopsy. Death results from acute asphyxia and is related to the rate and volume of blood loss and the underlying condition.[13,128,129,137,138] Factors increasing mortality include preexisting pulmonary insufficiency, obtundation from any cause, poor cough, and coagulopathy.[129,137,139,140]

Management consists of general measures, diagnosing the site of bleeding, isolating the bleeding lung, and controlling the bleeding. The patient should be given 100% O_2, placed in the head-down lateral-decubitus position, bleeding side down, clotting studies performed, blood typed, wide-bore intravenous access established, and cough suppressed with an opiate, and resuscitation and suction equipment should be in close proximity. Chest radiography and computed tomography scanning should be done when appropriate.

A number of alternatives exist for the localization, isolation, and control of the bleeding:

1. Fiberoptic bronchoscopy and placement of an endobronchial blocker (Fogarty catheter or Arndt endobronchial blocker) in the bleeding lung or segment should be done.[25,26,141–143] This can be performed in a patient who is not intubated but is easier in one who is. Fiberoptic bronchoscopy is limited by the narrow suction channel and rapid visual loss secondary to occlusion of the lens by blood.[140] Fiberoptic bronchoscopy, however, can allow accurate localization of bleeding,[104,128,140,144,145] catheter placement in a bronchial segment,[128,141,146] and lavage with iced isotonic saline or epinephrine.[128,146]

2. Rigid bronchoscopy has advantages over fiberoptic bronchoscopy when blood loss is massive because of better suction, visual access, and airway control.[137,139,140,144,147,148] It allows easier placement of endobronchial blockers and iced saline or epinephrine lavage. It also allows diathermy, laser therapy, and cryotherapy,[130] as well as placement of endobronchial tampons soaked in vasoconstrictor drugs.[137] As use of embolization increases, rigid bronchoscopy is required less, although it still has a place.[144,147]

3. Endotracheal intubation may be required where bleeding is sufficient to compromise oxygenation, particularly if mentation is depressed or cough is inadequate. If bleeding is so rapid that acute asphyxia arrest is imminent despite intubation,[128] the ETT can be advanced beyond the carina (usually into the right main bronchus) and OLV commenced.[128,149] If blood does not flow out of the ETT (implying blood loss from the contralateral side), the cuff is inflated and the ETT left in situ. If bleeding continues through the ETT, a Fogarty or Foley catheter is passed, the main bronchus is occluded, and the ETT is withdrawn to the trachea to ventilate the contralateral side.

4. Selective intubation is performed with a small (6 to 7 mm) ETT, with or without fiberoptic bronchoscope guidance, or a selective left or right endobronchial tube.[128,137] Selective intubation must be preceded by accurate bronchoscopic localization of bleeding because it excludes one lung from ventilation. Selective intubation has the advantage of reliable protection of the nonbleeding lung[137,140] but the disadvantage of permitting OLV only and excluding the bleeding lung from endobronchial procedures and suctioning. Bleeding then must be controlled by tamponade, bronchial angiography, and embolization or surgery.

5. DLT insertion is an alternative that isolates the lungs but preserves access to both. In the past, DLTs were not recommended as an early alternative[128,137] to control bleeding. Problems included difficulties with insertion under adverse circumstances, requirement for an experienced operator, difficulties with suction and bronchoscopy access, and easy blockage of DLT lumen with clot. More recently, use of PVC DLTs has been successful.[13,131,150] DLT enables lateralizing of the bleeding, protects the nonbleeding lung, allows ILV, and allows therapeutic procedures to address the bleeding lung without compromise of the healthy lung.[13,150] A left DLT is the tube of choice.[13,131,137,150] Most commonly, a single ventilator with a Y connection to the DLT lumens is used[13] with distribution of ventilation according to the relative impedance of the two lungs (see “Lung Separation” above). If there is risk of blood overflowing to the nonbleeding lung, or if differential ventilation is required, a second ventilator with AILV should be used. OLV must be present during bronchial blockade or endobronchial intubation and may be required during iced saline lavage or massive blood loss.

6. Embolization. Because 90% of major hemoptyses arise from bronchial arteries,[134] once bleeding has been localized, bronchial artery embolization has a high success rate.[104,134,145,151–153] Once the bleeding lung or segment has been isolated, the bleeding must be controlled.

7. Emergency surgery. Urgent surgical resection is undertaken if bleeding overwhelms airway control or fails to respond to other measures. Resection is associated with a low mortality in many series.[104,129,140,144,152,154] It should not be unduly delayed if bleeding is not readily controlled.

The choice of these alternatives depends on the intubation status of the patient, the rate of bleeding, and whether the site of bleeding is known (Table 25-3). The choice is also affected by the availability of the technique and the skills of the personnel involved. In an unintubated patient who remains stable despite moderate hemoptysis, bleeding may settle with correction of clotting abnormalities and general measures (above) or angiographic embolization (Table 25-3). An obtunded patient with a decompensating respiratory or circulatory state requires intubation combined with a plan to confine the bleeding to one lung (Table 25-3). This can include a primary choice of specialized tubes, such as the Univent or a DLT. A patient who is already intubated will usually have control attempted primarily by a blocker or manipulation of the ETT in situ (Table 25-3).

Table 25-3: Factors Influencing Choice of Procedures in Patients with Moderate to Massive Hemoptysis

	Intubation	Severity	Side of Bleed	Management Options
A	Not required	Moderate		Conservative only
				Angiogram and embolization
				Elective intubation and blocker
B	Required	Moderate		ETT + blocker[a] ± bronchoscopic guidance
		or massive	Known	Univent tube ± bronchoscopic guidance or DLT
		Massive	Known left	ETT advanced into R main bronchus
		Massive	Unknown	DLT or ETT to right main ± blocker[a] and withdraw ETT[b]
C	Already present	Moderate		Blocker[a] ± bronchoscopic guidance
		Massive		ETT to right main ± blocker and withdraw ETT[b]

[a]In an emergency, the blocker may be any suitable device with and inflatable balloon—Arndt, Cohen, Swan-Ganz catheter, balloon-tipped cardiac pacing wire, Fogarty catheter.

[b]If the site of bleeding is known, directed balloon placement can be done before bronchoscopy. If the site of bleeding is unknown, bronchoscopy may be required first to guide blocker placement.

In reported series of patients with major hemoptysis, localization of bleeding is usually achieved by history, fiberoptic bronchoscopy, or radiology (chest radiograph, computed tomography scan, or angiography). Bleeding is controlled by conservative measures, embolization, or surgery.[104,145,152] The relative requirement for these three treatments varies widely,[104,145,152] depending on cause, amount of bleeding, and local expertise. Supportive care, correction of coagulopathy, and time as the only measure ranged from 13% to 87% of patients with major hemoptysis. Embolization was used in 7% to 51% with success rates of more than 80%. Surgery was required in 6% to 50%.

Lung Protection from Secretions

Spread of purulent secretions from a lung abscess, empyema, bronchiectasis, cavitating tuberculosis, or cavitating malignant disease to the dependent normal lung during chest surgery is associated with considerable risk.[1,2,5,6,155] A DLT not only allowed thoracic surgery but provided an important protective role for the dependent normal lung[1,66,155] and allowed perioperative ILV if required.[1,66,155]

Although these conditions have become less common, the requirement for ILV for lung protection, during or outside thoracic surgery, occurs occasionally.[156] Essential during thoracic surgery, it is problematic outside that setting because viscous or tenacious secretions drain poorly through the narrower lumen of a DLT. Appropriate antibiotic therapy, postural drainage, and a standard ETT may be preferable.

Bronchial Repair Protection

Mainstem bronchi and lower trachea (near the carina) can be ruptured by severe blunt chest trauma, resulting in tension pneumothoraces, massive air leak, and the need for urgent surgical repair. Selective airway intubation usually is required for the repair and ILV has been used postoperatively to protect the anastomoses.[157–160] Pizov et al[160] used one-lung high-frequency ventilation in the management of traumatic tear of the bronchus in a child. After a right mainstem bronchial rupture, Moerer et al[157] used a left DLT with bilevel ventilation to the left lung and CPAP alone to the right lung for 48 hours before switching to CMV. In three patients with lower tracheal rupture (near the carina), Wichert[158] reported that standard DLTs position the cuff too close to the site of carinal injury and used bronchoscopy-guided selective endobronchial intubation with two tubes to undertake the repairs. After repair, the tubes were reintroduced via tracheostomy, and ILV was performed for 9 to 14 days to allow recovery from respiratory and other complications.

Bronchopleural Fistula

BPFs can result from trauma, necrotizing pneumonia, lung abscess, tuberculosis, acute respiratory distress syndrome (ARDS), thoracic surgery, overinflation from mechanical ventilation, central venous catheter insertion, and intercostal catheters. Persisting BPF often leads to infection of the pleural space.[161] If massive air leak from a BPF occurs during mechanical ventilation, the consequences include respiratory insufficiency and sometimes tension pneumothorax. Persisting failure of lung expansion can occur despite intercostal catheters if the rate of air leak exceeds the rate of drainage through the catheters. BPFs are estimated to occur in 2% of ventilated patients.[162] Mortality depends on the size of the leak[162] and the cause and is reported to be 18% to 67%.[161,162]

Management includes antibiotics, pleural sclerosing agents, surgical control of air leaks at thoracotomy, inter-costal catheter insertion, underwater seal drainage with suction, conventional ventilation strategies to reduce air leak, patient positioning,[161] positive pleural pressure during inspiration,[161,163] HFJV, a variety of bronchoscopic occlusion techniques, and ILV. ILV is usually employed when massive air leak persists despite conservative measures and results in respiratory insufficiency. Before instigating ILV, conservative measures should be optimized.

Inadequate drainage can occur despite actively bubbling intercostal catheters and can lead to failure of lung expansion and respiratory insufficiency. Intercostal catheters must be adequate in number and diameter; their patency must be visualized and demonstrated by active bubbling. Underwater-seal drainage systems and wall-suction units must have an adequate maximum flow capacity. Air-filter patency must be checked. High resistances and low maximum flow rates in either the drainage system or wall-suction unit actually can retard thoracic drainage and increase pneumothorax size. Ideally, maximum flow capacity should approximate or exceed the percentage of V_T lost to the BPF multiplied by the inspiratory flow rate. Drainage devices vary widely in maximum flow capacity but rarely exceed a capacity of 35 L/min. Increased bubbling or radiologic improvement after suction disconnection suggests retardation by the device. Persisting negative pressure on a wall-suction unit after disconnection suggests an occluded air filter. With massive air leaks, more than one underwater-seal drainage system may be required.

The conventional ventilator strategy for a BPF has three goals: reduce air-leak rate (to facilitate healing), reduce pneumothorax size, and maintain adequate gas exchange. These goals often have conflicting needs. The usual strategy is directed toward lowering alveolar and airway pressure to reduce air leak. V_T and PEEP should be minimized and the rate reduced, especially if dynamic hyperinflation is present, although these steps may impair gas exchange. Inspiratory flow is controversial. Increasing flow may decrease air leak by decreasing inspiratory time or increase proximal air leak by increasing peak airway pressure. The impact of any change on all three goals must be assessed.

HFJV without ILV appears to benefit patients with a proximal BPF and otherwise relatively normal lungs.[161,164] Reported success of HFJV in patients with parenchymal lung disease, whose BPFs usually are peripheral, is variable.[161] ILV has been used in many patients with a BPF.[35,47,56–58,60,62–64,66,88,162,164–174] Conditions in which ILV has been used for BPF include pneumonia with and without CAO,[174] ARDS,[168] trauma,[57,63,162,166,167] pulmonary contusion or large airway trauma,[62,133,164,166,171] emphysema, asthma,[165] staphylococcal pneumatoceles,[173] and thoracic surgery. [47,58,63,66,88] Most patients had air leaks exceeding 50% of V_T, lung collapse despite multiple intercostal catheter insertions, and hypercapnic acidosis and hypoxia despite attempts at optimizing mechanical ventilation.

The most common form of ILV for BPF is AILV using two ventilators[56–58,60,66] with low V_T, low rates, and low or no PEEP for the lung with the BPF. The BPF lung also has been ventilated with SILV with low V_T, PEEP, and inspiratory flow;[35,88,170–173] HFJV;[63,64] and high-frequency oscillation[168] or excluded from ventilation by a Fogarty catheter passed through a DLT after failing to respond to both CPAP and jet ventilation.[174] Successful use of a DLT with a single ventilator and a variable-resistance valve in the inspiratory circuit to the BPF lung has been reported in an animal model[48] and in one patient[47] with a large BPF.

Almost all authors report reduction in air leak with improvement in gas exchange. ILV was continued from 2 hours[66] up to 10 days[58] in some patients before CMV could be resumed. Improvement was reported in most patients with ILV. Overall survival was approximately 50%, although a large BPF is as a poor prognostic factor.[161] Outcome mainly was related to the prognosis of the underlying condition.

Asymmetrical Lung Disease

ILV has been used for a variety of unilateral or asymmetric lung diseases (see Table 25-1). Three main indications are unilateral pulmonary parenchymal injury, unilateral atelectasis, and unilateral airflow obstruction.

Unilateral Parenchymal Injury

Patients who receive ILV for asymmetric pulmonary injury have poor compliance on the affected side, hypoxemia refractory to high O_2 concentrations and high levels of PEEP. Under these circumstances, PEEP may cause hyperinflation of the unaffected lung, collapse of the affected lung,[41,50] barotrauma,[41,50,175,176] and worsening of hypoxemia[41] consequent to increased pulmonary vascular resistance in the unaffected lung (diverting blood flow to the injured lung). This hyperinflation also can elevate intrathoracic pressure, reduce arterial pressure and cardiac output,[41,50,169,170] and, combined with arterial desaturation, reduce O_2 delivery.[51,71,169,170,177–181]

The prime objective of ILV under these circumstances is differential PEEP, although different ventilator patterns have been applied to achieve a similar effect and optimize gas exchange and minimize barotrauma. ILV allows lung recruitment maneuvers and high PEEP to the affected lung, permitting maximum benefit to that lung without adverse effects on the contralateral lung, intrathoracic pressure, cardiac output, and the distribution of ventilation between the two lungs. PEEP applied to the diseased lung can improve oxygenation by alveolar recruitment and diverting blood flow to the more normal lung. Low or no PEEP in the more normal lung avoids hyperinflation and the adverse effects of high intrathoracic pressure.

ILV has been used in various unilateral or asymmetric lung diseases (see Table 26-1). The most common indication has been *pulmonary contusion.* Of forty-five patients who received ILV for asymmetric lung injury,[9,41,50,63,73,133,166,175,176,182,183] three received SILV,[41,50] but most received AILV.[9,166,175,176,182,184] Two patients received no mechanical ventilation and breathed spontaneously through a DLT with different levels of CPAP applied to the expired limb of each circuit.[73] All methods improved gas exchange, and overall mortality was only 10%. In twelve trauma patients with unilateral contusion requiring ILV, Cinnella et al[182] monitored end-tidal CO_2 and static compliance in each lung and reverted to conventional ventilation when these became similar in the two lungs.

ILV has been reported in patients with *aspiration*[55,56,169,170] and *pneumonia* or *consolidation.*[9,32,35,43,53,59,68,169,170,184–186] As with contusion, a mixture of AILV and SILV has been used. In one patient, high-frequency oscillatory ventilation was used with a higher mean airway pressure to the affected lung.[68] SILV has been used in patients with unilateral consolidation on a background of congenital heart disease with asymmetric lung blood supply.[32,185] Mortality was 48%.

SILV has been reported in patients with asymmetric ARDS secondary to trauma and sepsis (56% mortality[34]), asymmetric acute pulmonary edema,[32,169,187] and massive pulmonary embolism.[188] In a patient with a single-lung transplant, it enabled weaning from ECMO.[187]

All patients received differential PEEP: 0 to 10 cm H_2O on the unaffected side and 7 to 25 cm H_2O on the affected side. In all cases, higher PEEP was used on the affected side. V_T values to the two lungs were equal in some studies, smaller to the affected lung in some, and larger in one study.[50] In two studies,[34,169] PEEP administered to each affected lung was carefully adjusted to the compliance response of that lung. Carlon et al[169] increased PEEP in the affected lung until its compliance was similar to that of the unaffected lung. Siegal et al[34] found that increasing PEEP in the affected lung initially improved compliance, but then it decreased secondary to overinflation. They set PEEP at the level that achieved maximum compliance. Both methods improved oxygenation, shunt fraction, and cardiac output. The duration of ILV ranged from 1 hour[50] to 12 days.[176]

From these reports, several factors emerge as requirements for ILV:

1. *Differential PEEP* with a higher level applied to the affected lung is a key factor. PEEP may be applied to the affected lung until gas exchange improves, to an inflection point[34] on a compliance curve, until lung compliance is equal on the two sides,[169] or based on CO_2 excretion. PEEP most commonly is 10 to 20 cm H_2O. A recruitment maneuver and PEEP level that maximizes arterial oxygenation can be recommended.[189] PEEP may or may not be required in the contralateral lung.

2. *Equal* V_T to both lungs was used most commonly and was most likely to maximize gas exchange[36,39,40] compared to smaller or larger V_T values to the affected lung. Maintenance of plateau pressure below 30 cm H_2O is an important goal.[190,191]

3. *AILV* or *SILV* is equally acceptable because there is no requirement for a different respiratory rate. AILV holds no disadvantages when compared with SILV[55,61] and is simpler and more flexible.

The primary goal of ILV and differential PEEP or CPAP is improvement in gas exchange and hemodynamics and physical expansion of collapsed lung regions.

Unilateral Atelectasis

Unilateral atelectasis that has failed to respond to either standard ventilator support, bronchoscopy, or both[7,50,51,55,59,65,67,150,169,170,192–194] is another indication for ILV. High PEEP, CPAP, HFJV, or high-frequency oscillation have been applied to the atelectatic lung for the purpose of reinflation without the risk of overinflating the contralateral lung and generalized elevation of intrathoracic pressure.

In some studies,[42,51,59,67,169,170] high PEEP was applied to the atelectatic lung during AILV or SILV with mechanical ventilation of both lungs. The collapsed lung received 10 to 30 cm H_2O of PEEP, whereas the unaffected side received 0 to 10 cm H_2O of PEEP. In other studies, the collapsed lung received transient CPAP alone[7,65,67,150,193] (20 to 80 cm H_2O) without mechanical ventilation, or 30 cm H_2O CPAP followed by HFJV with 25 cm H_2O PEEP,[65] or 80 cm H_2O CPAP followed by AILV,[67] or unilateral high-frequency oscillation with a mean airway pressures of 28 to 30 cm H_2O.[192,194] Millen et al[193] transiently applied 60 to 70 cm H_2O of CPAP to individual lungs in two nonintubated, spontaneously breathing patients via a cuffed fiberoptic bronchoscope with good results. There was no report of lung injury despite transient application of very high inflation pressures, and there were only two deaths in this group of eighteen patients with atelectasis.[7,42,51,55,59,65,67,150,169,192,194]

These reports demonstrate success with a range of recruitment maneuvers. Application of CPAP up to 50 to 60 cm H_2O to both lungs for a few minutes has no prolonged consequences of raised intrathoracic pressure and is recommended for lung collapse.[189] In unilateral atelectasis, recruitment maneuvers should be undertaken routinely during double-lung ventilation (unless contraindicated by a problem in the nonatelectatic lung) before attempting ILV and may prevent the need for ILV in many patients.

Unilateral Airflow Obstruction

Unilateral airflow obstruction occurs most commonly following single lung transplantation (SLT) but may occur with a mechanical or chemical insult to one lung in a patient with asthma or partial occlusion of a major bronchus. Under these circumstances, standard ventilation can cause dynamic hyperinflation and high intrinsic PEEP in the obstructed lung. This can elevate intrathoracic pressure, reduce cardiac output, and compress the contralateral lung.

To achieve hypoventilation of the obstructed lung, SILV can be used with a much lower V_T to the obstructed lung. It is better achieved with AILV using reduced rate and V_T (or CPAP alone) to the obstructed lung.[54,67]

Single Lung Transplantation.

Early reports suggested that SLT was contraindicated in chronic obstructive pulmonary disease because of the risk of dynamic hyperinflation in the native lung with mediastinal shift.[122,195] Initial experience supported these concerns.[196–199] Subsequent series[10,195,200–209] combined with 210 SLTs at our institution[10] result in a total of 733 patients. Overall early mortality is 18% and only 13% when the primary diagnosis is CAO (emphysema, α_1-antitryspin deficiency, lymphangioleiomyomatosis) (Table 25-4). Thirty-day mortality varies widely (0%[203] to 50%[200]). Small series reveal that early mortality[195,200,207,208] ranges from being slightly lower with SLT than with bilateral lung transplant[195,207,208] to higher with SLT.[200] Larger series suggest lower mortality with bilateral lung transplantation,[210–213] although selection criteria such as age may have contributed.[210]

Table 25-4: Incidence of Acute Native Lung Hyperinflation and Incidence and Mortality with Independent Lung Ventilation in Patients Undergoing Single Lung Transplantation for Airflow Obstruction

	SLT for Airflow Obstruction			Radiologic ANLH		Symptomatic ANLH		ILV		ILV Mortality	
Authors	No. Pts	No. Died	Mortality	No.	%	No.	%	No.	%	No.	%
Kaiser et al[209]	11	0	0%	—		—		1	9%	0	0%
Patterson et al[195]	7	1	14%	1		1		0			
Egan et al[208]	4	0	0%	—		—		1	25%	0	0%
Marinelli et al[206]	7	1	14%	—		—		0			
Low et al[207]	16	2	13%	—		—		0			
Montoya et al[205]	39	1	3%	—		—		0			
Yonan et al[173]	27	5	19%	12	44%	12	44%	8	30%	2	25%
Weill et al[210]	51	0	0%	16	31%	8	16%	1	2%	0	0%
Mitchell et al[202]	132	34	26%	—		—		13	10%	6	46%
Hansen et al[201]	90	1	1%	—		—		0			
Angles et al[200]	14	7	50%	9	64%	9	64%	6	43%		

	SLT for Airflow Obstruction			Radiologic ANLH		Symptomatic ANLH		ILV		ILV Mortality	
Authors	**No. Pts**	**No. Died**	**Mortality**	**No.**	**%**	**No.**	**%**	**No.**	**%**	**No.**	**%**
Pilcher et al[10]	170	21	12%	78/95	82%	*20*	12%	20	12%	7	35%
Totals	**568**	**73**	**13%**	**116**	**60%**	**50**	**19%**	**50**	**9%**	**15**	**30%**

Abbreviations: ANLH, acute native lung hyperinflation; *ILV,* independent lung ventilation; *SLT,* single lung transplantation.

Although SLT is becoming less popular, it remains an alternative to bilateral lung transplantation, especially in patients older than 50 years of age with nonsuppurative lung disease. SLT is technically easier than bilateral lung transplantation and benefits more recipients when donor availability is a limiting factor. The most common indication for SLT is some form of CAO, which accounts for 77% of SLTs.[195,200–209] Although lung function is better after bilateral lung transplantation, SLT substantially improves quality of life[195,207] and achieves equivalent maximum work capacity and maximum O_2 consumption.[214]

More than thirty reports[10,27,54,67,187,195–209,215–226] plus experience from our institution provide information on 768 patients receiving SLTs, 601 for CAO. ILV was required almost exclusively in patients who received SLT for CAO.[10,27,54,200,202–204,208,209,215,217–219,221,223] There are few reports of patients requiring ILV after bilateral lung transplantation for any reason, nor after SLT for restrictive lung disease. In one patient,[223] ILV was required for a large, unresolving BPF arising in the native lung after SLT that eventually necessitated pneumonectomy of the native lung. In another patient,[187] ILV was required for reperfusion edema after SLT for primary pulmonary hypertension. Use of ILV in SLT series for CAO (see Table 25-4) varies widely: 0%[195,201,205–207] to 43%,[200] with an overall frequency of 9%.[10,195,200–209]

The phenomenon leading to use of ILV has been termed *acute native lung hyperinflation* (ANLH), which is defined as radiologic mediastinal shift with flattening of the ipsilateral hemidiaphragm (Fig. 25-8A) associated with signs of hemodynamic instability or respiratory dysfunction.[10,200,203,204] Evolution of this phenomenon can be divided into three stages:

1. *Asymptomatic ANLH.* Dynamic hyperinflation of the native lung with mediastinal shift is seen commonly in the postoperative period without transplanted lung collapse, hypoxia, or hypotension, and without the need for ILV.[10,195,225] This phenomenon arises because of asymmetry of lung disease following transplantation. The native lung with severe airflow obstruction undergoes dynamic hyperinflation during CMV, just as both lungs would do during CMV before transplantation.[227] A healthy transplanted lung with normal compliance and airflow resistance will receive most of the blood flow and ventilation, thereby reducing the degree of dynamic hyperinflation that

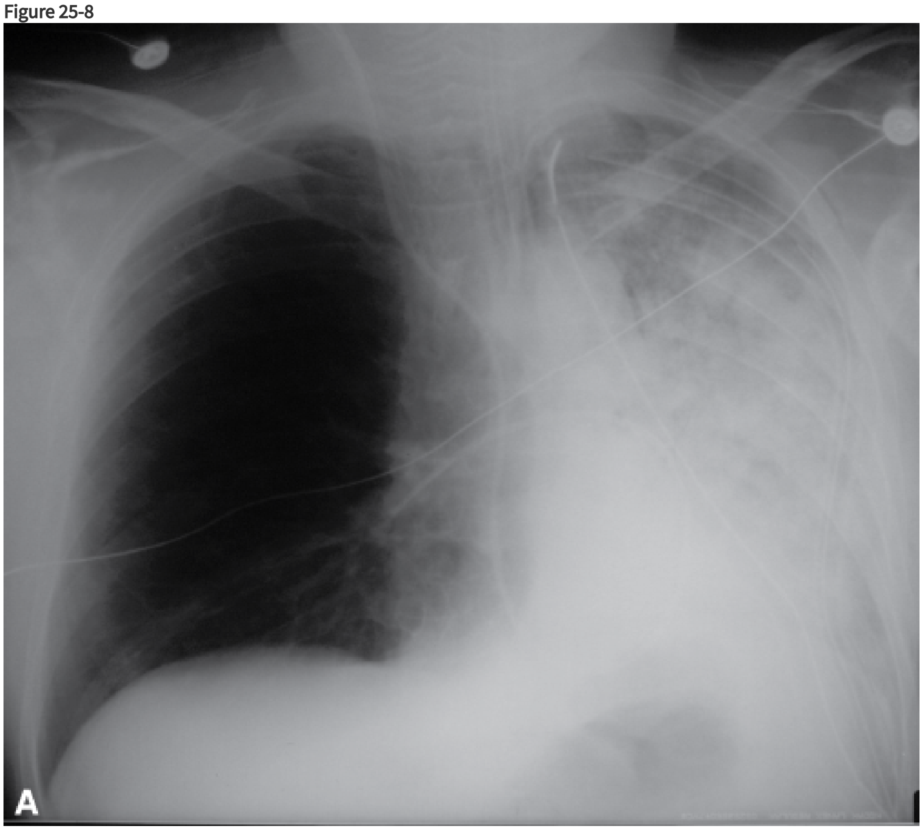
would have occurred if both lungs received the same ventilation. Nevertheless, the transplanted lung does not have a “balancing” degree of dynamic hyperinflation, and mediastinal shift occurs commonly. The occurrence of asymptomatic ANLH is reported infrequently (Table 25-5). Weill et al[203] reported asymptomatic ANLH in sixteen of fifty-one patients (31%), although smaller series have reported symptomatic ANLH in 44%[204] and 64%[200] of SLTs for CAO. In our institution, mediastinal shift (shift of right-heart border relative to the spine by 1 cm or greater toward the transplanted lung) occurred in seventy-eight of ninety-five consecutively evaluated SLTs (82%) for CAO (Table 25-5).[10] The asymmetry usually improves or resolves over time, but it persists indefinitely in some patients. In some patients it can arise weeks or months after transplant.[200]

2. *Transplanted lung dysfunction.* Any dysfunction in the transplanted lung, whether parenchymal (e.g., reperfusion edema, contusion, rejection, pneumonia, or collapse) or airway (e.g., anastomosis narrowing or sputum obstruction), impedes ventilation in the transplanted lung and redistributes ventilation to the native lung, especially during volume-controlled ventilation. This necessarily increases dynamic hyperinflation in the native lung and increases mediastinal shift and collapse and further impairs gas exchange in the transplanted lung. A vicious cycle results, redistributing more ventilation to the native lung with greater dynamic hyperinflation and mediastinal shift. Primary dysfunction of the transplanted lung,[219] severe postimplantation syndrome,[10,218] or ARDS have been identified as major contributors to native lung hyperinflation and mediastinal shift[10,27,208,209,218,219,221] usually resulting in hypotension, collapse of the transplanted lung with hypoxia, or both. Progressive compromise of the transplant lung can lead to refractory hypoxia and hypercapnia, whereas dynamic hyperinflation, mediastinal shift, and pulmonary vessel compression can lead to circulatory compromise with hypotension (see Fig. 25-8*A*). This has been termed *symptomatic ANLH*[203,204] (see Table 25-5). Almost all such problems have occurred only during mechanical ventilation, in the immediate or early postoperative period.

3. *Mechanical ventilation response.* The typical response to a lung collapse with worsening hypoxemia and hypercapnia includes increasing PEEP, increasing rate, and/or increasing V_T. Each response increases dynamic hyperinflation in the native lung,[227,228] which can worsen gas exchange, precipitate circulatory collapse, and necessitate ILV. Although the requirement for ILV usually is attributed to transplant dysfunction (see previous paragraph), the precipitant often is the failure to improve (or deterioration) with mechanical ventilation. In addition to transplanted lung injury (see previous paragraph), factors that increase the risk of this problem include the severity of airflow obstruction in the native lung, the size of the transplanted lung, and the side of the transplantation (see Table 25-5). Yonan et al[204] found that patients who developed ANLH had higher pulmonary artery pressures, higher residual volumes, and lower forced expiratory volumes in 1 second (FEV_1) than did patients who did not develop symptomatic ANLH. Weill et al[203] suspected more symptomatic ANLH in patients with bullous emphysema, but lung function and pulmonary artery pressure were equivalent among patients with and without symptomatic ANLH. We found that patients who required ILV had a greater degree of preoperative airflow limitation and preoperative hyperinflation, and lower postoperative Pa_{O_2}/FI_{O_2} ratios, more radiologic mediastinal shift, and more transplanted lung infiltrate on the postoperative chest radiograph.[10] Severity of airflow obstruction directly affects the degree of dynamic hyperinflation.[227,228] The size of the donor lung[195,209,218] is an important factor. A donor’s predicted vital capacity that approximates[209] or is smaller than[218] the recipient’s predicted vital capacity is associated with the need for ILV. A donor-to-recipient predicted vital capacity ratio of 1.4[209] or a donor-lung

predicted vital capacity exceeding recipient-lung predicted vital capacity by 2 L[195] is associated with the absence of mediastinal shift following SLT. Others have found no difference as a consequence donor size.[203]

Figure 25-8



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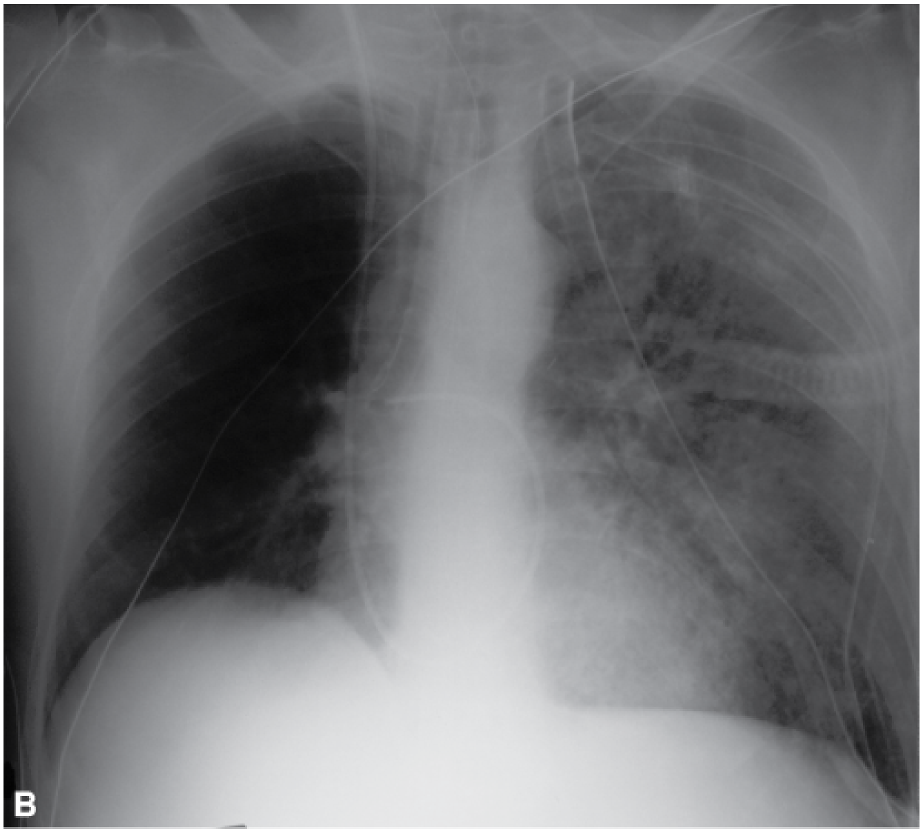

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Chest X-rays of a patient with a right single-lung transplant before (**A**) and (**B**) after insertion of a right double-lumen tube and independent lung ventilation.

Table 25-5: Incidence of Radiologic and Symptomatic Acute Native Lung Hyperinflation in Left and Right Single Lung Transplantation for Airflow Obstruction

	Radiologic ANLH						
	Left			Right			
	No.	**Total**	**%**	**No.**	**Total**	**%**	**P value**
Weill et al[179]	10	27	37	6	24	25	NS
Pilcher et al[10]	45	54	83	33	41	80	NS
Total	**55**	**81**	**68**	**39**	**65**	**60**	**NS**
	Symptomatic ANLH						
	Left			**Right**			
	No.	**Total**	**%**	**No.**	**Total**	**%**	**P value**
Weill et al[179]	4	27	15	4	24	17	NS
Angles et al[169]	5	6	83	4	8	50	NS
Pilcher et al[10]	14	82	17	7	88	8	NS
Total	**23**	**115**	**20**	**15**	**120**	**13**	**<0.05**

Abbreviations: ANLH, acute native lung hyperinflation; *NS,* nonsignificant; *SLT,* single lung transplantation.

Several reports have suggested a higher incidence of ANLH with left SLTs (see Table 25-5). Weill et al[203] found radiographic ANLH in 37% of left SLTs and 25% of right SLTs and no difference in symptomatic ANLH (see Table 25-4). Angles et al[200] did not report radiographic ANLH but found symptomatic ANLH in 83% of left SLTs and 50% of right SLTs. We[10] found no difference in mediastinal shift between left and right SLTs for CAO, but ILV was required twice as often with left SLTs (see Table 25-5).

Severe symptomatic ANLH commonly requires urgent intervention. Options include:

1. *Permissive hypercapnia.*[201] Permissive hypercapnia reduces dynamic hyperinflation in the native lung and may improve mild symptomatic ANLH but is unlikely to be adequate for severe ANLH. Although ILV was not used in this study, four patients required ECMO.

2. *ILV.*[27,54,198,200,202–204,208,209,215,217–219,221,223] Although technically complex, ILV commonly results in immediate resolution or improvement in gas exchange and circulatory problems and provides the most rapidly available and chosen solution (see Fig. 25-8).

3. *ECMO.*[201] ECMO is a complex and invasive solution, but provides an alternative to ILV.

4. *Contralateral lung-reduction surgery.*[202–204] This has been reported recently. It has been undertaken concurrently with SLT, during the postoperative period, and late after SLT. It may offer the best long-term solution to ANLH but is less readily applicable in urgent or life-threatening situations.

5. *Contralateral lung transplantation.*[202,204] After primary SLT, this offers a solution but depends on availability of a donor lung and subjects the recipient to two major surgical procedures and two sets of foreign antigens.

6. *Retransplantation of the SLT.*[204] This has been used for early graft dysfunction associated with ANLH.

The choice of DLT for ILV is different following SLT. A left DLT normally is easier to position correctly. With a left SLT, however, this can compromise the anastomosis and airway distal to the anastomosis, which has a tenuous blood supply in the immediate postoperative period. Thus, a DLT opposite to the side of the SLT normally is chosen in the early postoperative period. If ILV is required for more than 2 to 3 weeks after left SLT when the anastomosis is stable, a left DLT may be used, although a smaller size is chosen in case the anastomosis is narrower than the adjacent airway.

ILV after SLT usually consists of CMV to the transplanted lung and AILV,[208,209,218,219] SILV,[221] bronchial blockade,[27,209] CPAP,[223] or spontaneous ventilation[218] to the native lung. All methods are viable options. The ventilatory requirements of each lung differ so much, however, that AILV is the method of choice primarily because of different rate requirements of the two lungs (Table 25-6). AILV was the method of choice in forty-seven patients in four series.[10,200,202,204] The donor lung commonly requires PEEP at a sufficient level to expand collapsed regions and improve oxygenation, a V_T sufficiently low to avoid pressure injury to the lung, and a rate high enough for adequate CO_2 elimination without causing flow limitation. A recruitment maneuver may maximize the benefit from PEEP, provided that it is safe to undertake.[189,229,230] The native lung should be ventilated with a pattern that maximizes its contribution to gas exchange without excessive dynamic hyperinflation. This necessitates low V_T, high inspiratory flow rate, and a very low rate (Table 25-6).[228] The goals of native-lung ventilation are to maintain a low plateau pressure, restore the mediastinum to the midline (see Fig. 25-8B), and allow restoration of circulatory instability (that caused by dynamic hyperinflation). If this is not achieved, rate should be reduced further (even to 0). The goals of SLT ventilation are a safe plateau pressure (Table 25-7) and restoration of adequate SaO_2 and Pa_{CO_2}. If this is not achieved, further recruitment maneuvers and higher PEEP should be explored. The introduction and stabilization of these different ventilator strategies to each lung generally necessitate sedation and paralysis during the early stages of ILV. If SaO_2 or circulation is not satisfactory, ECMO may be needed.

Table 25-6: Ventilator Settings for Independent Ventilation after Single Lung Transplantation for Airflow Obstruction

	Native Lung	Transplanted Lung
Ventilator rate (breaths/min)	2 to 4	14 to 20
Tidal volume (mL/kg)	2 to 3	3 to 4
Plateau pressure (cm H_2O)	<25	<30
Inspiratory flow rate (L/min)	80	40 to 50
PEEP (cm H_2O)	0	10 to 17
Recruitment maneuver (cm H_2O)	No	±35 to 50[a]

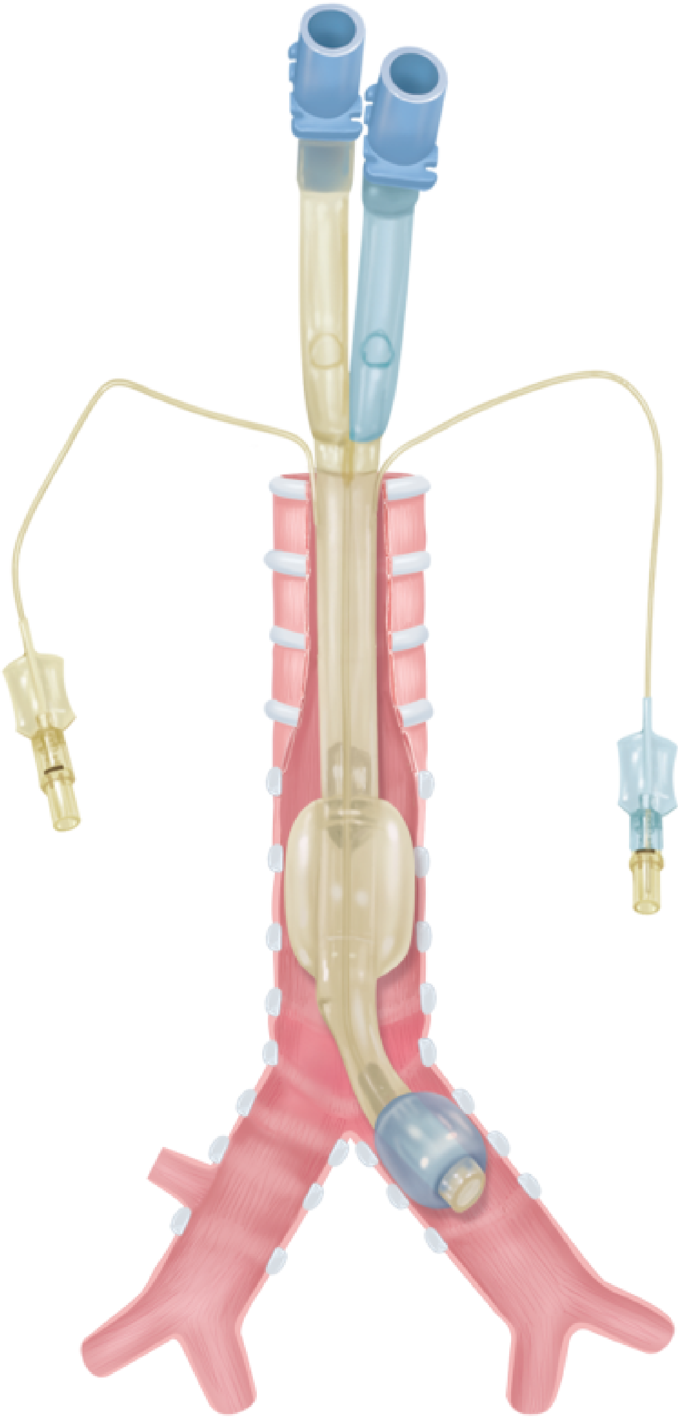
[a]Stability of the anastomosis and the presence of air leak need to be considered before undertaking a recruitment maneuver. Lower-than-normal recruitment pressures may be initially chosen.

Table 25-7: Complications Associated with Double-Lumen Tubes and Independent Ventilation

- Trauma to larynx, trachea, and major bronchi during insertion
- DLT malposition resulting in bronchial occlusion, poor oxygenation, ventilation, and secretion clearance
- Incomplete lung isolation resulting in unwanted lavage fluid, blood, or pus in the protected lung
- Retention of secretions as a result of long narrow airways and difficulty with suction
- Tube dislodgement following routine patient position changes
- Complications from prolonged immobility associated with the need for sedation and controlled ventilation
- Airway injury because of prolonged requirement for high cuff pressures to maintain
- Laryngeal injury from the prolonged presence of large diameter tube

Withdrawal of ILV can be initiated when function of the transplanted lung improves and its ventilatory requirements decrease. This can be assessed with a DLT in situ by ventilating both lungs with a single ventilator and a bifurcated ventilator circuit (see Fig. 25-3). Because spontaneous ventilation is difficult with a DLT, patients usually require reintubation with a single-lumen ETT for weaning before extubation. Others experience recurrent dynamic hyperinflation with double-lung ventilation and must wean with the DLT in situ. ILV has been required for as little as 1 day.[209] In most instances, prolonged ILV has been required; periods exceeding 1 month[218,219,221,223] have been reported with eventual resolution and good functional outcome. Patients who require prolonged ILV should undergo a tracheostomy and receive a double-lumen tracheostomy tube (Fig. 25-9).[218,221,231]

Figure 25-9



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A double-lumen tracheostomy tube.

Outcome of Acute Native Lung Hyperinflation.

Several investigators[10,200,203,204,218,221,231] have reported that patients with symptomatic ANLH have a longer duration of mechanical ventilation, a longer intensive care unit stay and higher mortality: 7% mortality in 213 SLT patients without symptomatic ANLH, and 37% mortality in fifty SLT patients with symptomatic ANLH ($P < 0.05$). Yonan et al[204] reported lower FEV1 values and higher residual volumes after transplantation in ANLH patients, whereas Weill et al,203 in a larger series, found no differences in long-term lung function.

Outcome of Independent Lung Ventilation.

Fifteen papers[10,27,54,198,200,202–204,208,209,215,217–219,221,223] report sixty-three patients with ILV following an SLT for CAO. Overall mortality was 30%. Including only recent case series in which all SLT patients and ILV survival were reported,[10,202–204] ILV patients had higher mortality (fifteen of forty-two patients, or 36% mortality) than non-ILV patients (forty-five of 338 patients, or 13% mortality). This difference is not surprising because ILV patients are sicker and show a higher incidence of graft dysfunction.

In summary, SLT for airflow obstruction poses a risk of ANLH with mediastinal shift. When this is sufficient to compromise gas exchange and/or circulation, ILV is the initial method of choice and may reduce the need for ECMO.

Unilateral Bronchospasm.

In patients with asthma, airways respond locally to local stimuli; irritants applied to one lung may create a different degree of bronchospasm in the two lungs.[232] During mechanical ventilation, this can lead to different levels of dynamic hyperinflation and mediastinal shift, and during thoracic surgery, this can lead to difficulty with thoracotomy closure and the need for ILV. Narr et al[67] reported unilateral bronchospasm during unilateral pleurodesis. The pleurodesed lung initially collapsed; following DLT insertion and ILV with 80 cm H_2O of CPAP applied to the collapsed lung, that lung inflated and would not deflate. Low-level CPAP to the affected lung, CMV to the unaffected lung, and aggressive bronchodilator therapy allowed deflation of the affected lung, maintenance of gas exchange, and thoracotomy closure.[67]

Patient Selection for Independent Lung Ventilation in Unilateral Lung Disease.

Asymmetric lung disease usually is easy to recognize, but criteria for ILV are not so clear. Not all patients with asymmetric lung disease require ILV. Commonly used criteria are unilateral or clearly asymmetric lung disease, which is evident on chest radiography or known from the patient's condition (e.g., SLT), plus one of the following:

1. Severe hypoxemia despite 100% O_2 and different levels of PEEP.
2. Circulatory failure/hypotension secondary to dynamic hyperinflation.

Factors that suggest that a patient is likely to benefit from ILV include

1. Worsening hypoxemia and/or circulatory status by increasing PEEP, rate, or V_T.
2. Improvement in gas exchange but marked deterioration in circulatory status with increasing PEEP and the opposite when PEEP is reduced.

Bilateral Symmetrical Lung Disease

Acute bilateral lung injury may appear diffuse, uniform, and symmetric on chest radiography but contain significant in homogeneity on computed tomography scan.[233,234] Many of these changes result from a uniformly injured lung collapsing in dependent zones as a result of increased lung weight. Because patients commonly are nursed on their back, collapse occurs in posterior zones and is not apparent on anteroposterior chest radiographs. In this case, no PEEP is required to open alveoli in the least dependent regions, and high PEEP may be required in the most dependent regions. Generalized PEEP may fail to inflate the most dependent regions and may overinflate nondependent regions. There is now increasing evidence that the injury can result from prolonged collapse[235] and repetitive collapse/reexpansion and overexpansion,[236,237] and that lung injury may contribute to multiorgan failure and death.[190,238] Considerable effort has been devoted to ventilator strategies that reduce prolonged collapse, collapse/reexpansion, and overexpansion,[190,191,239] and more recently to reduce prolonged collapse by higher PEEP with or without lung recruitment.[240–243] Although some strategies have been successful, all have the same problem of conflicting pressure requirements within different regions of the same lung.

The application of ILV under these circumstances is based on two principles:

1. Differences in compliance, ventilation–perfusion ratios, and gas exchange between the two lungs may be present and not be suspected from plain X-rays.[34,39,40,169]

2. Placing the patient in the lateral decubitus position will redistribute much of the collapsed (high-PEEP-requiring) lung regions to the dependent lung and the open (low-PEEP-requiring) regions to the nondependent lung.[39,40,244,245]

When a patient with normal lungs is ventilated in the lateral decubitus position, up to two-thirds of perfusion goes to the dependent lung, and up to two-thirds of ventilation goes to the nondependent lung,[36,75] creating significant ventilation–perfusion mismatch and shunt. Application of global PEEP improves distribution of ventilation,[246] but perfusion inequality may be increased and cardiac output reduced[36] with little net benefit.

In acute bilateral lung disease, the changes in ventilation and perfusion and the resulting mismatch in the lateral decubitus position are similar to those seen in patients with normal lungs[37,38] but with redistribution of lung collapse from posterior lung zones to the dependent lung. ILV with equal V_T to each lung and selective PEEP to the dependent lung proved additive in improving shunt fraction, arterial oxygenation, cardiac output, and O_2 delivery compared with CMV in both the supine and lateral positions.[37,38] These maneuvers have been applied on a long-term basis in bilateral lung injury with some early success.[245] Wickerts et al[244] and Diaz-Reganon Valverde et al[33] prospectively studied eleven and forty-five patients, respectively, with severe bilateral ARDS who received ILV. These patients were assessed in the supine position on global PEEP and then reintubated with a DLT and placed in the lateral decubitus position. PEEP was applied to the dependent lung and no PEEP to the nondependent lung. Both groups reported improved oxygenation, and Diaz-Reganon Valverde et al[33] reported a good response in 83% of patients.

ILV is difficult and labor-intensive to apply.[244] Accordingly, it has not gained widespread acceptance for symmetric bilateral lung injury.

Complications with Independent Lung Ventilation

Many of the problems with ILV are related to the DLT. The technique of DLT placement and verifying its correct position is complex, time-consuming, and requires experienced personnel.[15,19,21,23] These considerations can limit the use of DLTs and ILV when required urgently or under adverse circumstances, such as massive hemoptysis.[137,140]

Trauma of the larynx and upper airways was a common problem with red-rubber tubes[19,106] but is uncommon with PVC tubes.[15,247,248] Rare complications have included cardiovascular collapse after a DLT displaced a mediastinal tumor into the mediastinal vessels[249] and exsanguination following inclusion of the DLT in sutures during pneumonectomy with subsequent vessel laceration when the DLT was removed.[250,251]

Placement of right DLTs is critical with respect to right upper-lobe ventilation. Benumof et al[252] estimated a right upper-lobe occlusion rate of 11% with PVC tubes. McKenna et al[18] assessed occlusion of the right upper lobe by bronchoscopy: occlusion was 89% with Mallinckrodt PVC DLTs and only 10% with right Robertshaw red-rubber DLTs. High malposition rates are reported after primary insertion.[22,253] This has become an uncommon problem now that most DLT insertions are assessed routinely by bronchoscopy. Left upper-lobe occlusion also has been reported[254,255] with left PVC DLTs from overinsertion, wedging the tip of the DLT in the left lower-lobe bronchus. Saito et al[256] found 27 ± 6 mm movement of the tip of a left DLT, proximally with neck extension and distally with flexion, with a total potential range of movement of up to 4.5 to 7 cm. This range of movement is sufficient to either occlude the left upper-lobe bronchus or lose lung isolation. Similar problems are more likely with a right DLT, with which tube position is more critical.

High cuff pressures are a significant problem with DLTs. In routinely placed tubes with cuffs inflated to avoid air leak, bronchial cuff pressures were 56 ± 21 mm Hg in left PVC DLTs and 130 ± 41 mm Hg in Carlens tubes.[14,257] This study suggests a considerably lower risk to the airway from PVC tubes, although the pressures required to prevent leaks still were well above safe limits.

Bronchial rupture was reported regularly[248,258–263] and there is one report of bronchial stenosis[183] with red-rubber tubes. Bronchial rupture is uncommon with PVC DLTs, although they are not free of this complication.[248,250,264]

Although lumen diameter has improved considerably with PVC tubes, difficult suction access, retained secretions,[41,43,60,88] and lumen blockage remain a problem and can lead to difficult ventilation, the need for regular bronchoscopic toilet, and the need to change the DLT.[13]

When ILV is continued for some time, several difficulties related to patient care arise. Head movement, patient movement, and routine patient turning all threaten DLT position and can lead to loss of lung isolation and lobe occlusion.[43,60,88,256] Frequent bronchoscopy may be required to maintain DLT position.[43,248] Running two ventilators requires additional space, O_2, air, and suction outlets, and standard patient charts usually are inadequate.[60] There are significant increases in nursing time requirement and workload,[60,265,266] and the patient may find the DLT more uncomfortable and restrictive than normal intubation.[88]

Conclusion

ILV has been used in a wide range of conditions (see Table 25-2). Although ILV is infrequently required, it has an established and sometimes lifesaving role for many of conditions and users of mechanical ventilation should be familiar with the complexities of ILV so that it can be instigated promptly and appropriately when the need arises. A ILV can be undertaken with any ventilator and can be used for all indications.

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